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HEAT EXCHANGERS INCLUDING PARTIAL HEIGHT FINS HAVING AT LEAST PARTIALLY FREE TERMINAL EDGES

Abstract

In an embodiment, a heat exchanger includes a monolithic body that includes a first substrate, a second substrate, a third substrate, and a plurality of partial height fins. The second substrate is arranged parallel to and spaced from the first substrate, thereby defining a first fluid flow path. The third substrate is arranged parallel to and spaced from the second substrate opposite the first substrate, thereby defining a second fluid flow path. The plurality of partial height fins extend from one of the second substrate and the third substrate toward the other of the second substrate or the third substrate, wherein a terminal edge of each partial height fin is at least partially spaced from the other of the second substrate or the third substrate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] The present application is a continuation of co-pending U.S. application Ser. No. 18/582,960 filed Feb. 21, 2024, which is a division of U.S. patent application Ser. No. 17/223,290, filed Apr. 6, 2021, now U.S. Pat. No. 11,940,232 issued Mar. 26, 2024, entitled “Heat Exchangers Including Partial Height Fins Having At Least Partially Free Terminal Edges,” the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present specification generally relates to heat exchangers.

BACKGROUND

[0003] Heat exchangers may be conventionally produced through stacked plates that include etched, milled, or stamped grooves, which provide for fluid flow passages. Instead of stacked plates it may be desirable to have a more monolithic design. Monolithic designs may, however, face increased thermal strains as walls and grooves are formed as a single, interconnected piece.

[0004] Accordingly, a need exists for alternative monolithic heat exchangers with improved performance under thermal stress.

SUMMARY

[0005] In an embodiment, a heat exchanger includes a monolithic body that includes a first substrate, a second substrate, a third substrate, and a plurality of partial height fins. The second substrate is arranged parallel to and spaced from the first substrate, thereby defining a first fluid flow path. The third substrate is arranged parallel to and spaced from the second substrate opposite the first substrate, thereby defining a second fluid flow path. The plurality of partial height fins extend from one of the second substrate and the third substrate toward the other of the second substrate or the third substrate, wherein a terminal edge of each partial height fin is at least partially spaced from the other of the second substrate or the third substrate.

[0006] In another embodiment, a heat exchanger includes a monolithic body including a plurality of substantially parallel substrates layered one over the other, and a plurality of partial height fins extending from one of the plurality of substrates toward an adjacent one of the plurality of substrates. A terminal edge of each fin is only partially spaced from the adjacent one of the plurality of substrates to define a connecting foot that is connected to the adjacent one of the plurality of substrates.

[0007] In yet another embodiment, a heat exchanger includes a monolithic body including a first substrate; a second substrate, a third substrate, and a plurality of partial height fins. The second substrate is arranged parallel to and spaced from the first substrate, thereby defining a first fluid flow path. The third substrate is arranged parallel to and spaced from the second substrate opposite the first substrate, thereby defining a second fluid flow path. The plurality of partial height fins extend between the second substrate and the third substrate, wherein a terminal edge of each fin is at least partially spaced from one of the second substrate or the third substrate. The plurality of partial height fins include a first type of partial height fin extending from the second substrate into the second fluid flow path, a second type of partial height fin extending from the third substrate into the second fluid flow path, and a plurality of connector fins. The plurality of partial height fins include a plurality of rows of partial height fins arranged side-by-side, wherein each row includes

fins of the first type and the second type. The plurality of connector fins each connect a first fin in a first row to a second fin in a second row, wherein each connector fin is arranged at an oblique angle to a flow direction through the second fluid flow path.

[0008] These and additional features provided by the embodiments described herein will be more fully understood in view of the following detailed description, in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The embodiments set forth in the drawings are illustrative and exemplary in nature and not intended to limit the subject matter defined by the claims. The following detailed description of the illustrative embodiments can be understood when read in conjunction with the following drawings, where like structure is indicated with like reference numerals and in which:

[0010] FIG. 1A schematically depicts a side view of a heat exchanger having a plurality of partial height fins, according to one or more embodiments shown and described herein;

[0011] FIG. 1B schematically depicts a cross-sectional view of the heat exchanger of FIG. 1A taken along line 1B-1B, according to one or more embodiments shown and described herein;

[0012] FIG. 1C depicts a cross-sectional view of the heat exchanger of FIG. 1A taken along line 1C-1C, according to one or more embodiments shown and described herein;

[0013] FIG. 2A schematically depicts another embodiment of a heat exchanger having a plurality of partial height fins, according to one or more embodiments shown and described herein;

[0014] FIG. 2B depicts a side view of the heat exchanger of FIG. 2A, according to one or more embodiments shown and described herein;

[0015] FIG. 3A schematically depicts a cross-section view of another embodiment of a heat exchanger, according to one or more embodiments shown and described herein;

[0016] FIG. 3B depicts a front view of the heat exchanger of FIG. 3A, according to one or more embodiments shown and described herein;

[0017] FIG. 3C depicts a top view of the cross-section of the heat exchanger of FIG. 3A, according to one or more embodiments shown and described herein;

[0018] FIG. 4A depicts supporting and non-supporting substrate layers for use as layers of a heat exchanger having a non-monolithic structure, according to one or more embodiments shown and described herein;

[0019] FIG. 4B depicts a perspective view of a heat exchanger assembled from the supporting and non-supporting substrate layers of FIG. 4A, according to one or more embodiments shown and described herein; and

[0020] FIG. 4C depicts a side view of the heat exchanger of FIG. 4B, according to one or more embodiments shown and described herein.

DETAILED DESCRIPTION

[0021] Embodiments of the present disclosure are directed to heat exchangers that include partial height fins having at least partially free terminal edges. The heat exchangers may include a monolithic body comprising a first substrate, a second substrate, a third substrate, and a plurality of partial height fins. The fins are termed “partial height” in that they at least partially terminate free at terminal edges leaving a gap between the terminal edges and the closest adjacent substrate. The second substrate is arranged parallel to and spaced from the first substrate, thereby defining a first fluid flow path. The third substrate is arranged parallel to and spaced from the second substrate opposite the first substrate, thereby defining a second fluid flow path. The plurality of partial height fins extend between the second substrate and the third substrate, wherein a terminal edge of each fin is at least partially spaced from one of the second substrate or the third substrate. As will be described in greater detail herein, use of partial height fins may provide improved thermal

performance, reduced thermal strain, and longer heat exchanger life.

[0022] Heat exchangers are devices that are used to transfer heat between two or more fluids. Heat exchangers may be used for engine cooling (e.g., in the aviation or vehicle industries), electronics cooling, or the like. In particular, heat exchangers according to the present disclosure may be useful in high-pressure environments (e.g., up to and/or including pressures of 10,000 psi or greater). Referring generally to the figures, various monolithic heat exchanger embodiments are depicted. As used herein, the term “monolithic” refers to a structure formed as a single piece, such as through use of an additive manufacturing process, though other manufacturing processes are contemplated and possible.

[0023] For example, heat exchangers, according to the present disclosure, may be additively manufactured by forming layers or substrates having a plurality of partial height fins extending therebetween. That is, the heat exchangers may be integrally formed such that each layer is integrally connected to the proceeding layer through such processes as additive manufacturing or printing, metal casting, machining, etc. Additive printing techniques include, for example, selective laser sintering (SLS), direct metal laser sintering (DMLS), and other three dimensional printing (3DP) modalities. The materials can include stainless steel, aluminum, titanium, Inconel 625, Inconel 718, cobalt chrome, among other metal materials. In addition, ceramics may be used for very high-temperature applications. In each of these powder-based fabrication methods, powdered material is melted or sintered to form each part layer. For example, the SLS process utilizes powdered plastic materials that are selectively sintered by a laser layer-by-layer. Other types of additive manufacturing techniques include 3D printing including stereolithography (SLA), jetted photopolymer, or ink jet printing. Other types of additive printing include solid-based processes, which use non-powdered materials that are layered one on top of another and subsequently cut out. These methods includes laminated object manufacturing (LOM) or fused deposition modeling (FDM). Any of the above techniques may be utilized to form the integral heat exchangers of the present disclosure. However, such monolithic designs may result in increased thermal strains relative to conventional plate layers which may then be coupled to one another together. Accordingly, and as will be described in greater detail herein, embodiments of the present disclosure include partial height fins, which may introduce a level of thermal compliance that improves thermal performance of the heat exchanger to reduce thermal stress, thereby increasing to life of the heat exchanger.

[0024] As used herein, the term “partial fin height” or “partial height fin” refers to a fin, which has a terminal edge that is at least partially spaced from a substrate that is opposite the substrate from which it extends. That is, at least a part of a partial height fin extends only a portion of the distance between two parallel substrates. In this way, a gap may be positioned between partial height fin at the terminal edge and the substrate that is opposite the substrate from which it extends, thereby providing increased flexibility and thermal compliance to the heat-exchanger substrate.

[0025] Referring now to FIGS. 1A-1C, a heat exchanger **100** is schematically depicted. The heat exchanger **100** includes a body **102** which is monolithically formed by one or more of the methods described above, such that each layer of the body **102** of the heat exchanger **100** is integral with the next. It is noted that in some embodiments, each layer may be separately formed and then fixed to one another via brazing, welding, diffusion bonding, or the like, as will be described in further embodiments below.

[0026] Still referring to FIGS. 1A-1C, the monolithic body **102** may include a plurality of substantially parallel substrates such as a first substrate **110**, a second substrate **120** arranged parallel to and spaced from the first substrate **110** in the Z direction of the depicted coordinate axes, thereby defining a first fluid flow path **104** in the X direction. A third substrate **130** may be arranged parallel to and spaced from the second substrate **120** in the Z direction of the depicted coordinate axes, opposite the first substrate **110**, thereby defining a second fluid flow path **106** in the Y direction. In some embodiments, a fourth substrate **150** may be arranged parallel to and

spaced from the third substrate **130**, in the Z direction of the depicted coordinate axes, opposite the second substrate **120**, thereby defining a third fluid flow path **108** in the X direction.

[0027] In embodiments, the first fluid flow path **104** and the second flow path flow **106** may be arranged in a cross-flow pattern with one another such that a direction of flow **105** of the first fluid flow path **104** is perpendicular to the direction of flow **107** (depicted in FIG. **1B**) of the second fluid flow path **106**. In other embodiments, the direction of flow **105** of the first fluid flow path **104** may be parallel to the direction of flow **107** of the second fluid flow path **106**. Other orientations are contemplated and possible. For example, the flow paths **104**, **106**, **108** may not be substantially linear, as depicted, but may include turns or curves. Additionally, the third fluid flow path **108** may be arranged such that a direction of flow **109** through the third fluid flow path **108** is parallel to the direction of flow **105** through first fluid flow path **104**. In other embodiments, the direction of flow **109** through the third fluid flow path **108** may instead be parallel to the direction of flow **107** through the second fluid flow path **106**.

[0028] In embodiments, the body **102** of the heat exchanger **100** includes a first sidewall **103a** and a second sidewall **103b** (FIG. **1A**) positioned on either side of the fluid flow paths **104**, **106**, and/or **108**. The first sidewall **103a** and the second sidewall **103b** may extend between and connect the first substrate **110**, the second substrate **120**, the third substrate **130**, and/or the fourth substrate **150**. Openings (not shown) formed within the sidewalls **103a**, **103b** may provide access for fluidically coupling the fluid flow paths **104** and **108** through tubing, manifolds, or the like, allowing fluid to be supplied to multiple passages through a common interface.

[0029] Arranged within the second fluid flow path **106**, between the first sidewall **103a** and the second side wall may be a plurality of partial height fins **140**. Each fin **140** of the plurality of partial height fins **140** may extend generally parallel to the vertical direction (i.e., the Z-axis of the depicted coordinate axes). Each fin **140** may be sized and shaped to minimize thermal stresses between the fins **140** and the opposite substrate from which it extends. For example, each fin **140** may have a thickness of about 5 mm to about 50 mm along the X direction of the depicted coordinate axes and may have a length (along the Y direction) and a height (along the Z direction) that are larger than the width-wise dimension of the fin **140**. The lengthwise dimension of the plurality of fins may be arranged parallel to the flow direction **107**, as illustrated in FIG. **1B**.

[0030] The plurality of partial height fins **140** may extend between the second substrate **120** and the third substrate **130**. As described above, partial height fins refer to fins which extend only a portion of the distance between the second substrate **120** and the third substrate **130**. For example, each fin **140** may include a terminal edge **144**, the terminal edge **144** of each fin **140** is the edge facing opposite the second substrate **120** or the third substrate **130** from which the fin **140** extends and is at least partially spaced from one of the second substrate **120** or the third substrate **130**. At an end opposite the terminal edge **144** the fins are integrally formed with the second or third substrate **120**, **130**. As noted above, each find **140** is sized and shaped to minimize thermal stresses between the plurality of fins and the opposite substrate from which is extends. This reduction in thermal stress is provided via minimizing coupling between the terminal edge **144** and the opposite substrate. For example, the terminal edge **144** may be partially or completely uncoupled from the opposite substrate. Accordingly, the height of the partial height fin **140** where the terminal edge **144** of the fin **140** may extend less than 100% of the distance between the second substrate **120** and the this substrate **130**, such as between about 20% and about 99%, such as between about 50% and about 98%, so other percentage distances are contemplated and possible.

[0031] In embodiments, the heat exchanger **100** may include a first type **141a** of partial height fins **140**, which extend from the second substrate **120** into the second fluid flow path **106** (e.g., toward the third substrate **130**) and a second type **141b** of partial height fins **140**, which extend from the third substrate **130** into the second fluid flow path **106** (e.g., toward the second substrate **120**). As may be observed in FIG. **1B**, the plurality of partial height fins **140** may include a plurality of rows **146a-146f** extending along the X direction of the depicted coordinate axes, wherein each row is

arranged side-by-side to one another in the Y direction (e.g., the longitudinal direction) of the depicted coordinate axes. Each row **146a-146f** may alternate from including fins **140** of the first type **141a** in one row to fins of the second type **141b** in an adjacent row. Stated another way, and as depicted in FIG. **1C**, the first row **146a** may only include fins **140** of the second type **141b** (e.g., extending from the third substrate **130**, wherein at least a portion of the terminal edge **144** of each fin **140** is spaced from the second substrate **120**), a second row **146b** may include only fins **140** of the first type **141a** (e.g., extending from the second substrate **120**, wherein at least a portion of the terminal edge **144** of each fin **140** is spaced from the third substrate **130**), a third row **146c** may include only fins **140** of the second type **141b**, a fourth row **146d** may include only fins **140** of the first type **141a**, and so one and so forth. It is noted that the heat exchanger **100** may include any number of rows without departing from the scope of the present disclosure (e.g., two or more rows, three or more rows, four or more row, etc.).

[0032] Referring to FIGS. **1A** and **1C**, the partial height fins **140** of each row **146a-146f** may be laterally offset in the X direction of the depicted coordinate axes from the partial height fins **140** of an adjacent row. Accordingly, when viewed along the Y direction of the depicted coordinate axes, fins **140** of the first type **141a** appear to be located between fins **140** of the second type **141b** in the X direction of the depicted coordinate axes. Staggering the partial height fins **140** in the X restarts thermal boundary layers and introduces additional fluid mixing, which can increase heat transfer efficiency.

[0033] Using partial height fins **140** provides several advantages. For instance, use of partial height fins **140** structurally decouples the heat exchanger **100** thermal performance from the stress/life of the heat exchanger **100**. In particular, partial height fins **140** reduce mechanical coupling between the second substrate **120** and the third substrate **130** compared to fully coupled fins, which may reduce thermal stress experienced by the heat exchanger **100**. Moreover, thermal compliance of monolithic heat exchangers are improved as the partial height fins **140** provide flexure to allow for thermal expansion/contraction. Accordingly, use of partial height fins **140** may improve life of the monolithic heat exchanger **100** leading to cost savings and a reduced need for maintenance. Additionally, partial height fins **140** within the overall heat exchanger structure may be difficult to manufacture using traditional manufacturing techniques. With additive manufacturing procedures, partial height fins **140** may be more easily achieved and may be provided with various modifications, as will be described in further detail herein, by simply modifying the computer aided design. In embodiments, partial height fins **140** may be designed to enhance heat transfer coefficient while reducing pressure drop relative to a conventional fin (e.g., full-height fin). Accordingly, monolithic heat exchangers may be provided as a competitive alternative to conventional heat exchangers. Moreover, conventional plate-fin heat exchangers may be practically limited to operating pressures of less than 400 psia. For conventional heat exchangers to be operated at pressures about 400 psia, fin thickness required for mechanical strength may offset thermal performance. However, high temperature differences introduce thermal strains, independent of operating pressures, and partial height fins as described herein may be advantageous in such applications (e.g. having high differential temperature operation).

[0034] Still referring to FIGS. **1A-1C**, in embodiments, e.g., either in the first fluid flow path **104**, the third fluid flow path **108**, or both, one or more stiffening elements **160** (e.g., connecting walls) may extend between and connect the first substrate **110** and the second substrate **120** and/or the third substrate **130** and the fourth substrate **150**. For example, a plurality of stiffening elements **160** (e.g., two or more, three or more, fourth or more, etc.) may extend between the first sidewall **103a** and the second sidewall **103b**. The plurality of stiffening elements **160** may extend along the first and third fluid flow paths **104**, **108** in the X direction of the depicted coordinate axes. In some embodiments, each of the plurality of stiffening elements **160** may extend across the entire heat exchanger **100** in the X direction of the depicted coordinate axes forming separated flow paths that are subsets of the overall flow path **104**, **108**. In some embodiments, the plurality of stiffening

elements **160** may extend across only a portion of the heat exchange in the X direction of the depicted coordinate axes. As depicted in FIG. 1B, each of the stiffening elements **160** may be positioned so as to be aligned with a break between each row of the plurality of rows **146a-146f**. The plurality of stiffening elements **160** may provide additionally rigidity to the structure.

[0035] As an example use case, the heat exchanger **100** may be used in an air-cooled oil cooler (ACOC) exchanger. In such embodiments, high pressure fluid streams of oil may be directed through the first fluid flow path **104** and/or the third fluid flow path **108**. Heat from the hot oil may transfer to the heat exchanger **100**. Cool air (or other cooling fluid) hitting the plurality of partial height fins **140** may absorb and remove heat from the heat exchanger **100**. The plurality of partial height fins **140** may provide increased surface area (e.g., more exposed edges as compared to standard fins due to the terminal edges **144** and the staggered arrangement of rows) to allow the cool air to draw heat from the heat exchanger **100**. The additional surface area may improve heat exchange efficiency. Moreover, as the partial height fins **140** are at least partially uncoupled from the second substrate **120** or the third substrate **130**, the partial height fins **140** may bend or flex in the X direction of the depicted coordinate axes, which may reduce thermal stress, which may otherwise be experienced by more rigidly constrained fins. It is noted that in some embodiments, the heat exchanger **100** may be used as a heat source heat exchanger to add heat to a system as opposed to a heat sink heat exchanger to remove heat.

[0036] As noted herein, the heat exchanger **100** may be manufactured using additive manufacturing according to any of the techniques described herein. In embodiments, the heat exchanger **100** may be built along any direction such as along the X direction, the Y direction or the Z direction of the depicted coordinate axes. However, in some embodiments, the heat exchanger **100** may be formed through any conventional manufacturing process, which may include brazing the fins on each substrate and then brazing, welding, and/or diffusion bonding each fin/substrate layer into a final stacked heat exchanger configuration. Alternatively, the fins and/or substrates may be fixed in place (e.g., via a fixture) and the entire assembly brazed together in a single operation.

[0037] Embodiments below provide alternative embodiments of the heat exchanger **100** which are substantially similar to the embodiment described above unless otherwise noted or apparent from the figures. As will be described below, the following embodiments, may provide some additional benefits to improve manufacturability of monolithic heat exchangers using additive manufacturing.

[0038] FIGS. 2A and 2B depict an alternative embodiment of a heat exchanger **200**. The heat exchanger **200** is similar to the heat exchanger **100** in that the heat exchanger **200** includes partial height fins and cross-flow paths among other things. Accordingly, the above-description of the heat exchanger **100** may be applicable to the embodiments of FIGS. 2A and 2B unless otherwise noted or apparent from the figures. In the depicted embodiment, each fin **240** of the plurality of fins comprises a convex longitudinal leading edge **215** and a concave longitudinal trailing edge **217**. That is, the leading edge **215** of the fin **240** in the longitudinal direction (i.e., the Y direction of the depicted coordinate axes) is angled or curved convexly toward the -Y direction, and the trailing edge **217** of the fin **240** in the longitudinal direction is angled or curved concavely in the -Y direction of the depicted coordinate axes. Additionally, each stiffening element **260** located in either of a first fluid flow path **204** or a third fluid flow path **209** may be obliquely angled relative to the first substrate **210** and second substrate **220** (or the third substrate **230** and the fourth substrate **250**) in the Y-Z plane of the depicted coordinate axes.

[0039] The angled surfaces of the convex longitudinal leading edge **215**, the concave longitudinal trailing edge **217**, and the stiffening elements **260** may provide improved arrangement to support additive manufacturing, such as through DMLM. In particular, and with reference to FIG. 2B, the heat exchanger **200** is arranged on a build platform **10** the build direction **12** is arranged vertically. It may be difficult to manufacture purely horizontal walls (e.g., walls arranged perpendicular to the build direction **12** when performing additive manufacturing, particularly horizontal walls suspended at a position spaced from the build platform **10**. Accordingly, the angled and/or curved

surfaces of the convex longitudinal leading edge **215**, the concave longitudinal trailing edge **217**, and the stiffening elements **260** can facilitate manufacturing by orienting the walls at an angle that is oblique to the horizontal axis making the walls more self-supporting. In embodiments, the angled surfaces of the convex longitudinal leading edge **215**, the concave longitudinal trailing edge **217**, and the stiffening elements **260** may be angled between about 10 degrees and 80 degrees relative to the horizontal (e.g., the ground or build platform), such as between about 40 degrees and 60 degrees relative to the horizontal, such as about 45 degrees.

[0040] In some embodiments, it is contemplated that the leading edge **215** of the fin **240** may be flat or parallel with the horizontal while the trailing edge **217** of the fin **240** is concave. Such embodiments may still allow for case of manufacturing with additive manufacturing, but may result in fins having a reduced heat transfer area. Accordingly, a convex longitudinal leading edge **215** may preserve heat transfer area, thereby providing improved heat transfer properties.

[0041] To maintain the profile of each fin **240** during manufacturing, it may be desirable to tie each fin **240** to both the second substrate **220** and the third substrate **230**. For example, each of the partial height fins **240** may be connected to an opposite substrate from which the partial height fins **240** extends via one or more connecting feet **205a**, **205b**. For example, and as illustrated, each fin **240** may connect to an opposite substrate via a first connecting foot **205a** and a second connecting foot **205b**. The first connecting foot **205a** may be arranged at the convex longitudinal leading edge **215** and the second connecting foot **205b** may be arranged at the concave longitudinal trailing edge **217**. Accordingly, the terminal edge **244** of each fin **240** may be spaced from the opposite substrate from which it extends between the first connecting foot **205a** and the second connecting foot **205b** so as to be at least partially spaced from the one of the second substrate **220** or the third substrate **130**.

[0042] While the one or more connecting feet **205a**, **205b** may provide additional stability during manufacture, the one or more connecting feet **205a**, **205b** may be unnecessary during use of the heat exchanger **200**. Accordingly, in embodiments, during use of the heat exchanger **200**, the one or more connecting feet **205a**, **205b** may be designed to be breakable under the stresses experienced by the heat exchanger **200** during use. In such embodiments, a rib **208** of increased local material thickness may extend from the second substrate **220** and/or the third substrate **230** where the one or more connecting feet **205a**, **205b** connect thereto. The rib **208** may prevent damage to the heat exchanger **100** (e.g., openings between the first fluid flow path **204** or the third fluid flow path **209** to the second fluid flow path **206**) when the one or more connecting feet **205a**, **205b** break from the second substrate **220** or the third substrate **230**. The rib **208** may extend across the entire heat exchanger **100** in the X direction of the depicted coordinate axes, or only a portion thereof.

[0043] Referring now to FIGS. 3A-3C yet another embodiment of a heat exchanger **300** is schematically depicted. The heat exchanger **300** is similar to the heat exchanger **100** embodiments described above. For example, the heat exchanger **300** includes partial height fins **340** positioned within a fluid flow path **306**. Accordingly the above-description of the heat exchanger **100/200** is applicable to the embodiment of FIGS. 3A-3C unless otherwise noted or apparent from the figures and will not be repeated. While not shown, the heat exchanger **300** may include a plurality of fluid flow paths, beyond fluid flow path **306**, such as illustrated with respect to heat exchangers **100/200**, described above. For example, the heat exchanger **300** may include a first fluid flow path, a second fluid flow path, a third fluid flow path etc., to provide any cross-flow or counter-flow heat exchanger as desired.

[0044] In the present embodiment, positioned within the fluid flow path **306** in conjunction with the plurality of partial height fins **340** may be a plurality of connector fins **302**. The plurality of connector fins **302** may each connect a first fin **340a** of one row to a second fin **340b** in a second row adjacent to the first row.

[0045] In this embodiment, as with embodiments above, the plurality of partial height fins **340** are arranged in a plurality of rows **346a-346f**. However, in this embodiment, each row includes fins

340 of both the first type **341a** (e.g., extending from substrate **320**, wherein at least a portion of the terminal edge **344** of each fin **340** is spaced from the opposite substrate **330**), and the second type **341b** (e.g., extending from substrate **330**, wherein at least a portion of the terminal edge **344** of each fin **340** is spaced from the opposite substrate **320**). For example, the first row **346a** may include fins **340** of the first type **341a** alternating with fins **340** of the second type **341b** such that every other fin **340** is a fin **340** of the first type **341a** and every remaining fin **340** is a second type **341b**. The second row **346b** may also include fins **340** of the first type **341a** alternating with fins **340** of the second type **341b** such that every other fin **340** is a first type **341a** and every remaining fin is a second type **341b**, but in opposite order to the first row **346a**. The third row **346c** may be arranged substantially similar to the first row **346a**, the fourth row **346d** may be arranged substantially similarly to the second row **346b**, and so on depending on the number of desired rows. [0046] In embodiments, the plurality of connector fins **302** may connect the fins of each rows **346a-346f**. For example, a connector fin **302** may connect a first fin **340a** of the first row **346a** to a second fin **340b** in the second row **346b** adjacent to the first row **346a**. In embodiments, the plurality of connector fins **302** may connect two fins **340** of the same type (e.g., first type **341a** or second type **341b**), such as illustrated in FIG. 3C. In embodiments, connector fins **302** may connect the first fin of each row to one another, the second fins of each row to one another, the third fins of each row to one another, etc. In some embodiments, and as illustrated in FIG. 3C, a connector fin **302** may connect a fin **340** to a sidewall (e.g., the second sidewall **303b** opposite the first sidewall **303a** in the present embodiment) where there is not a corresponding fin **340** in the adjacent row. [0047] It is noted that the connector fins **302**, similar to the plurality of partial height fins **340** described above, may extend from one of substrate **320** and substrate **330** and be at least partially (or wholly) spaced from the other of substrate **320** and substrate **330** depending on which substrate **320**, **330** that the connector fin **302** extends from. For example, where a connector fin **302** is connecting fins **340** of the first type **341a**, the connector fin **302** may extend from substrate **320** and may be spaced from substrate **330**. Similarly, wherein a connector fin **302** is connecting fins **340** of the second type **341b**, the connector fin **302** may extend from substrate **330** and may be spaced from substrate **320**. Each of the connector fins **302** may be arranged at an oblique angle relative to the direction of flow **307** through the fluid flow path **306** in the X-Z plane of the depicted coordinate axes in FIG. 3C.

[0048] Still referring to FIGS. 3A-3C, in embodiments, the plurality of connector fins **302** may each define one or more fluid openings **304** extending therethrough and arranged to allow fluid to flow through the plurality of connector fins **302** along the direction of flow **307** of the fluid flow path **306**. In some embodiments, each connector fin **302** may define a plurality of fluid openings **304** extending therethrough (e.g., two or more, three or more, four or more, etc.). The size and/or shape of the openings may be designed to provide certain desired flow properties through the fluid flow path **306**. For example, though the one or more fluid openings **304** are depicted as oval-shaped, the one or more fluid openings **304** may be any polygonal or non-polygonal, regular or irregular shape (e.g., circular, rectangular, triangular, etc.). In embodiments, it is contemplated the one or more fluid openings **304** may provide balanced flow through the second fluid flow path **106**. For example, an increased number of fluid openings **304** at a smaller diameter may balance flow split between the various channels defined by the plurality of partial height fins **340** and connector fins **302**.

[0049] Referring to FIG. 3C the heat exchanger **300** is arranged on a build platform **10** and the build direction **12** is arranged vertically (along the Z direction of the depicted coordinate axes). As noted above, it may be difficult to manufacture horizontal surfaces (e.g., surfaces arranged perpendicular to the build direction **12** when performing additive manufacturing, particularly surfaces suspended at a position spaced from the build platform **10**). Accordingly, by obliquely angling each connector fin **302**, the heat exchanger **300** may be more easily manufactured by additive manufacturing the connector fins **302** may provide support to support a leading or trailing

edge of each fin **340**, such as described above. In embodiments, the angled orientation of each connector fin **340** relative to the direction of flow **307** through the fluid flow path **306** may be between about 10 degrees and 80 degrees relative to the horizontal, such as between about 40 degrees and 60 degrees relative to the horizontal, such as about 45 degrees.

[0050] As noted above, in some embodiments, instead of forming the heater exchanger monolithically, it may be desirable to form separate layers, which may provide for simplified mechanical analysis and more determinist load paths. In particular, FIG. **4A** illustrates an alternative embodiment including separate substrate layers for production of a heat exchanger **100**, illustrated in FIGS. **4B** and **4C**. In this embodiment, the heat exchanger **400** may be formed from one or more supporting substrate layers **410** layered with one or more non-supporting substrates layers **440**. Each of the substrate layers **410**, **440** may be additively manufactured and diffusion bonded or otherwise coupled to one another. However, other manufacturing processes are also contemplated and possible.

[0051] The supporting substrate layer **410** may include a base plate **411** having a first outer surface **412** and a second outer surface **414** opposite the first outer surface **412**. The first and second outer surfaces **412**, **414** may be parallel and coextensive with one another. A sidewall **416** may extend between the first outer surface **412** and the second outer surface **414**. The sidewall **416** may be perpendicular to both the first outer surface **412** and the second outer surface **414** and extend along an outer perimeter of the first outer surface **412** and the second outer surface **414**. For example, where the overall shape of the supporting substrate layer **410** is rectangular such that there may be four sidewalls.

[0052] A first plurality of fins **420** may be formed (e.g., through additive manufacturing) and extend from the first outer surface **412** of the base plate **411**. Each fin may extend generally parallel to the vertical direction (i.e., the Z-axis of the depicted coordinate axes). Each fin of the first plurality of fins **420** may have a thickness of about 5 mm to about 10 mm along the Y direction of the depicted coordinate axes and may have a length (along the X direction) and a height (along the Z direction) that is larger than the width-wise dimension of the fin. The lengthwise dimension of the plurality of fins **420** may be arranged parallel to the flow direction **415**, as illustrated.

[0053] The first plurality of fins **420** may be arranged in two or more discrete rows of fins (e.g., row **422a**, **422b**, and/or **422c**, though a greater number of rows are contemplated and possible) that extend in a line transverse to the flow direction **415** of fluid through the first plurality of fins **420** (i.e., along the Y direction of the depicted coordinate axes). Each row **422a**, **422b**, and/or **422c** may be arranged parallel to an adjacent row and extends across the substrate along the Y-axis of the depicted coordinate axes.

[0054] Two or more attachment walls **426** may be formed and extend from the first outer surface **412** of the base plate **411** to a distance (i.e., height) greater than the first plurality of fins **420**. The two or more attachment walls **426** may thereby provide attachment points at which the non-supporting substrate layer **440** may be bonded (e.g., through diffusion bonding) or otherwise coupled to the non-supporting substrate layer **440** without contacting the first plurality of fins **420**. The two or more attachment walls **426** include at least a first attachment wall **426a** and a second attachment wall **426b**. The first and second attachments walls **426a**, **426b** may be arranged at either end of the first plurality of fins **420** in the Y direction of the depicted coordinate axes such that the first plurality of fins **420** are arranged completely between the first attachment wall **426a** and the second attachment wall **426b**. In some embodiments, and as illustrated, additional attachment walls may be arranged between the first attachment wall **426a** and the second attachment wall **426b**, such as a third attachment wall **426c** and/or a fourth attachment wall **426d**. The third attachment wall **426c** and/or the fourth attachment wall **426d** may divide the first plurality of fins **420** into two or more discrete groups (e.g., group **423a**, group **423b**, and/or group **423c**). Each group **423a**, **423b**, **423c** may have an equal number of fins or an unequal number of fins.

[0055] The two or more attachment walls **426** may have a thickness in the Y direction of the

depicted coordinate axes that is greater than the thickness of the fins of the first plurality of fins **420**. In yet further embodiments, the thickness of the two or more attachment walls **426** may be less than or equal to the thickness of the first plurality of fins **420**. It is noted that while the two or more attachment walls **426** are illustrated as having a uniform thickness along their lengths in the X direction of the depicted coordinate axes, in some embodiments, the two or more attachment walls **426** may have more truss-like frames such that openings are formed through the thicknesses of the two or more attachment walls. Such openings may provide for desirable weight reduction. In some embodiments, only the internally arranged attachment wall(s) (e.g., third attachment wall **426c** and/or fourth attachment wall **426d**) may include openings formed therethrough, while the first and second attachment walls **426a**, **426b** are solid without any openings formed therein, to prevent fluid from leaking from the heat exchanger.

[0056] Formed in the second outer surface **414** of the base plate **411** may be one or more partial fluid flow channels **418**. The one or more partial fluid flow channels **418** may extend along the entire length of the second outer surface **414** in the Y direction of the depicted coordinate axes. The one or more partial fluid flow channels **418** may have any cross-sectional shape, for example, a portion of a circle, oval, rectangle, or any regular or irregular polygonal or non-polygonal shape. In the illustrated embodiment, the one or more partial fluid flow channels **418** may include a plurality of partial fluid flow channels **418** formed within the second outer surface **414**, for example, two or more partial fluid flow channels, four or more partial fluid flow channels, six or more partial fluid flow channels, etc. Each of the one or more partial fluid flow channels **418** may run parallel to each other across the base plate **411**. Each of the one or more partial fluid flow channels **418** may have the same shape or different shapes. As will be described in greater detail, the one or more partial fluid flow channels **418** of the supporting substrate layer **410** may be combined with one or more partial fluid flow channels **446** formed in a non-supporting substrate layer **440** to form one or more combined fluid flow channels **448**, as illustrated in FIG. 4B.

[0057] The non-supporting substrate layer **440** may include a base plate **441** having a first outer surface **442** and a second outer surface **443** opposite the first outer surface **442**. The first and second outer surfaces **442**, **443** may be parallel and coextensive with one another. A sidewall **444** may extend between the first outer surface **442** and the second outer surface **443**. The sidewall **444** may be perpendicular to both the first outer surface **442** and the second outer surface **443** and extend along an outer perimeter of the first outer surface **442** and the second outer surface **443**. For example, where the overall shape of the non-supporting substrate layer **440** is rectangular, there may be four sidewalls.

[0058] A second plurality of fins **450** may be formed (e.g., through additive manufacturing) and extend from the first outer surface **442** of the base plate **441**. Each fin may extend generally parallel to the vertical direction (i.e., the Z-axis of the depicted coordinate axes). Each of the fins of the second plurality of fins **450** may have a thickness of about 2 mm to about 10 mm along the Y direction of the depicted coordinate axes and may have a length (along the X direction) and a height (along the Z direction) that are larger than the width-wise dimension of the fin. The lengthwise dimension of the second plurality of fins **450** may be arranged parallel to the flow direction **415**.

[0059] The second plurality of fins **450** may be arranged in two or more discrete rows of fins (e.g., row **450a**, **450b**, and/or **450c**, through a greater number of rows are contemplated and possible) that extend in a line transverse to the flow direction **415** of fluid through the second plurality of fins **450** (i.e., along the Y direction of the depicted coordinate axes). Each row **450a**, **450b**, and/or **450c** may be arranged parallel to an adjacent row and extends across the substrate along the Y-axis of the depicted coordinate axes.

[0060] Each row may be separated into two or more groups (e.g., group **452a**, group **452b**, and/or group **452c**). The two or more groups may be apparent due to an increased spacing (e.g., s.sub.1, s.sub.2) between adjacent groups as compared to spacing between adjacent fins within a single

group. The increased spacing may allow for insertion of the attachment walls **426c** and/or **426d** to extend between and separate the second plurality of fins into the two or more groups, as illustrated in FIG. **4C**.

[0061] Formed in the second outer surface **443** of the base plate **441** may be one or more partial fluid flow channels **446**. The one or more partial fluid flow channels **446** may extend along the entire length of the second outer surface **443** in the Y direction of the depicted coordinate axes. The one or more partial fluid flow channels **446** may have any cross-sectional shape, for example, a portion of a circle, oval, rectangle, or any regular or irregular polygonal or non-polygonal shape. In the illustrated embodiment, the one or more partial fluid flow channels **446** may include a plurality of partial fluid flow channels **446** formed within the second outer surface **443**, for example, two or more partial fluid flow channels, four or more partial fluid flow channels, six or more partial fluid flow channels, etc. Each of the one or more partial fluid flow channels **446** may run parallel to each other across the substrate. Each of the one or more partial fluid flow channels **446** may have the same shape or different shapes. As will be described in greater detail, the one or more partial fluid flow channels **446** of the non-supporting substrate layer **440** may be combined with one or more partial fluid flow channels **418** formed in a supporting substrate layer **410** to form one or more combined fluid flow channels **448**, as illustrated in FIG. **4B**.

[0062] It is further noted, that though not shown, the supporting and non-supporting substrate layers **410**, **440** may include one or more alignment mechanisms to aid in aligning the one or more partial fluid flow channels **418**, **446**. For example, a protrusion may be formed on one of the second outer surface **414** of the supporting substrate layer **410** or the second outer surface **443** of the non-supporting substrate layer **440** and a matching recess may be formed on the other of the second outer surface **414**, **443** of the supporting substrate layer **410** or the non-supporting substrate layer **440**. During assembly, the protrusion may be mated with the recess to aid in alignment and assembly.

[0063] In some embodiments, when forming the supporting and non-supporting substrate layers **410**, **440** with additive manufacturing (e.g., DMLM) the direction of build may be along the Z direction of the depicted coordinate axes. This direction may optimize fin formation so as to produce fins having thinner profiles (e.g., less than about 20 mm thick, less than about 15 mm thick, less than 12 mm thick, between about 5 mm to about 20 mm thick, such as about 15 mm to about 20 mm, or the like). However, it is noted that other build directions are contemplated and possible (e.g., along the Y direction and/or along the X direction). Additionally, it is noted that the supporting substrate layer **410** and the non-supporting substrate layer **440** may be formed through other manufacturing processes (e.g., mechanically coupling fins to a surface of a substrate and etching, milling, and/or carving partial fluid flow channels on an opposite surface) such as described herein.

[0064] During the assembly, the non-supporting substrate layer **440** may be rotated about the X axis of the depicted coordinate axes and placed on the two or more attachment walls **426** of the supporting substrate layer **410**. As illustrated in FIGS. **4B** and **4C**, a heat exchanger **400** as depicted is formed from alternating layers of supporting and non-supporting substrate layers **410**, **440**. It is noted that a heat exchanger **400** includes at least a first supporting substrate layer **410a** and a first non-supporting substrate layer **440a**. When assembled, the first outer surface **442** of the non-supporting substrate layer **440** sits on the two or more attachment walls **426** (e.g., **426a-426d**) of the supporting substrate layer **410**. Each adjacent fin of the first plurality of fins **420** are thereby separated from one another by a fin of the second plurality of fins **450**. In this way, one or more fluid flow paths are formed by the first plurality of fins **420**, the second plurality of fins **450**, and the two or more attachment walls **426**. Additionally, as shown, the first plurality of fins **420** and the second plurality of fins **450** are partial height fins in that they are at least partially spaced from the opposite substrate layer from which they extend, thereby providing greater thermal compliance, improved life, and reduced stress.

[0065] As illustrated in FIGS. 4B and 4C, a subsequent supporting substrate layer **410b** may be layered on top of the second outer surface **443** of the non-supporting substrate layer **440a**, in the Z direction of the depicted coordinate axes, such that the plurality of partial fluid flow channels **418** of the subsequent supporting substrate layer **410b** and the plurality of partial fluid flow channels **446** of the non-supporting substrate layer **440** are combined to form a plurality of combined fluid flow channels **448** that define a plurality of fluid flow paths. The flow direction **449** of the combined fluid flow channels **448** may be parallel to or transverse to the flow direction **415** of the one or more fluid flow paths provided by the first plurality of fins **420**, the second plurality of fins **450**, and the two or more attachment walls **426** (as illustrated in FIG. 4B). Additional non-supporting substrate layers (e.g., **440b**, **440c**) and supporting substrate layers (**410c**) may be added as desired to form the desired heat exchanger **400**. The various layers may be diffusion bonded or otherwise coupled to one another. Prior to bonding the various layers together, each substrate layer may be inspected for defects, geometrical tolerances, and/or hydraulic and/or thermal performance characteristics.

[0066] It is noted that in the above embodiments, though the fin are illustrated as having a particular configuration, it is contemplated different rows of the heat exchanger may have different configurations. For example, in some embodiments, each layer may have partial height fins as described herein. In other embodiments, only some layers may have partial height fins. Moreover, in some embodiments, fins within the same layer may include partial height fins and full height fins. For example, it may be advantageous to have partial height fins within select regions of a heat exchanger, while other regions of the heat exchanger include full height fins.

[0067] Further aspects of the present disclosure are provided by the subject matter of the following numbered clauses. [0068] 1. A heat exchanger comprising: a monolithic body comprising: a first substrate; a second substrate arranged parallel to and spaced from the first substrate, thereby defining a first fluid flow path; a third substrate arranged parallel to and spaced from the second substrate opposite the first substrate, thereby defining a second fluid flow path; and a plurality of partial height fins extending from one of the second substrate and the third substrate toward the other of the second substrate or the third substrate, wherein a terminal edge of each partial height fin is at least partially spaced from the other of the second substrate or the third substrate. [0069] 2. The heat exchanger of clause 1, wherein the plurality of partial height fins are sized and shaped to minimize thermal stresses between each fin and an opposite substrate from which it extends. [0070] 3. The heat exchanger of any preceding clause, further comprising a plurality of stiffening elements extending between the first substrate and the second substrate. [0071] 4. The heat exchanger of any preceding clause, further comprising a fourth substrate arranged parallel to and spaced from the third substrate opposite the second substrate, thereby defining a third fluid flow path. [0072] 5. The heat exchanger of any preceding clause, wherein the plurality of partial height fins comprise: a first type of partial height fin extending from the second substrate into the second fluid flow path; and a second type of partial height fin extending from the third substrate into the second fluid flow path. [0073] 6. The heat exchanger of any preceding clause, wherein the plurality of partial height fins comprise a plurality of rows of partial height fins arranged side-by-side in a longitudinal direction wherein each row alternates between the first type of partial height fin and the second type of partial height fin. [0074] 7. The heat exchanger of any preceding clause, wherein the plurality of partial height fins of at least one row is laterally offset in a direction perpendicular to the second fluid flow path from the plurality of partial height fins of an adjacent row. [0075] 8. The heat exchanger of any preceding clause, wherein the plurality of partial height fins is connected to an opposite substrate from which the plurality of partial height fins extends via one or more connecting feet. [0076] 9. The heat exchanger of any preceding clause, wherein each fin comprises a convex longitudinal leading edge and a concave longitudinal trailing edge. [0077] 10. A heat exchanger comprising: a monolithic body comprising: a plurality of substantially parallel substrates layered one over the other; and a plurality of partial height fins extending from one of the plurality

of substrates toward an adjacent one of the plurality of substrates, wherein a terminal edge of each fin is only partially spaced from the adjacent one of the plurality of substrates to define a connecting foot that is connected to the adjacent one of the plurality of substrates. [0078] 11. The heat exchanger of any preceding clause, further comprising a plurality of stiffening elements extending between a first substrate and a second substrate of the plurality of substrates, wherein each stiffening element is obliquely angled relative to the first substrate and the second substrate. [0079] 12. The heat exchanger of any preceding clause, wherein each fin comprises a convex longitudinal leading edge and a concave longitudinal trailing edge. [0080] 13. The heat exchanger of any preceding clause, wherein the plurality of partial height fins comprise: a first type of partial height fin, wherein the first type extends from a first substrate of the plurality of substrates into a fluid flow path; and a second type of partial height fin, wherein the second type extends from a second substrate of the plurality of substrates into the fluid flow path. [0081] 14. The heat exchanger of any preceding clause, wherein the plurality of partial height fins comprise a plurality of rows of partial height fins arranged side-by-side in a longitudinal direction wherein each row alternates between the first type of partial height fins and the second type of partial height fins. [0082] 15. The heat exchanger of any preceding clause, wherein the plurality of partial height fins of at least one row is laterally offset in a direction perpendicular to the fluid flow path from the plurality of partial height fins of an adjacent row. [0083] 16. The heat exchanger of any preceding clause, wherein the plurality of partial height fins are connected to an opposite substrate from which the plurality of partial height fins extends via a first connecting foot and a second connecting foot. [0084] 17. A heat exchanger comprising: a monolithic body comprising: a first substrate; a second substrate arranged parallel to and spaced from the first substrate, thereby defining a first fluid flow path; a third substrate arranged parallel to and spaced from the second substrate opposite the first substrate, thereby defining a second fluid flow path; and a plurality of partial height fins extending between the second substrate and the third substrate, wherein a terminal edge of each fin is at least partially spaced from one of the second substrate or the third substrate, the plurality of partial height fins comprising: a first type of partial height fin extending from the second substrate into the second fluid flow path; a second type of partial height fin extending from the third substrate into the second fluid flow path, wherein the plurality of partial height fins comprise a plurality of rows of partial height fins arranged in side-by-side, wherein each row comprises fins of the first type and the second type; and a plurality of connector fins each connecting a first fin in a first row to a second fin in a second row, wherein each connector fin is arranged at an oblique angle to a flow direction through the second fluid flow path. [0085] 18. The heat exchanger of any preceding clause, further comprising a plurality of stiffening elements extending between the first substrate and the second substrate, wherein each stiffening element is obliquely angled relative to the first substrate and the second substrate. [0086] 19. The heat exchanger of any preceding clause, further comprising a fourth substrate arranged parallel to and spaced from the third substrate opposite the second substrate, thereby defining a third fluid flow path. [0087] 20. The heat exchanger of any preceding clause, wherein each connector fin defines one or more fluid openings extending therethrough. [0088] 21. The heat exchanger of any preceding clause, wherein every other fin in each row of the plurality of rows of partial height fins is a fin of the first type, and a connector fin of the plurality of connector fins connects two fins of the first type. [0089] It should now be understood that embodiments as described herein are directed to heat exchangers that include partial height fins. For example, in at least some embodiments, heat exchangers according to the present disclosure include a monolithic body comprising a first substrate, a second substrate, a third substrate, and a plurality of partial height fins. The second substrate is arranged parallel to and spaced from the first substrate, thereby defining a first fluid flow path. The third substrate is arranged parallel to and spaced from the second substrate opposite the first substrate, thereby defining a second fluid flow path. The plurality of partial height fins extend between the second substrate and the third substrate, wherein a terminal edge of each partial

height fin is at least partially spaced from the other of the second substrate or the third substrate. Use of partial height fins (as opposed to full-height fins) may provide improved thermal performance, reduced thermal strain, and longer heat exchanger life.

[0090] Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about”, “approximately”, and “substantially”, are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 1, 2, 4, 10, 15, or 20 percent margin. These approximating margins may apply to a single value, either or both endpoints defining numerical ranges, and/or the margin for ranges between endpoints

[0091] While particular embodiments have been illustrated and described herein, it should be understood that various other changes and modifications may be made without departing from the scope of the claimed subject matter. Moreover, although various aspects of the claimed subject matter have been described herein, such aspects need not be utilized in combination. It is therefore intended that the appended claims cover all such changes and modifications that are within the scope of the claimed subject matter.

Claims

1. A method of manufacturing a heat exchanger comprising: integrally forming a monolithic body by: forming a first substrate; forming a second substrate, wherein a fluid flow path is defined between the first substrate and the second substrate; and forming a plurality of partial height fins extending between the first substrate and the second substrate, the plurality of partial height fins comprising: a first type of partial height fin extending from the first substrate into the fluid flow path; and a second type of partial height fin extending from the second substrate into the fluid flow path, wherein the plurality of partial height fins comprise a plurality of rows of partial height fins arranged side-by-side consecutively along the fluid flow path, wherein fins of each row are offset from fins of an adjacent row in a direction perpendicular to a flow direction along the fluid flow path.
2. The method of claim 1, wherein integrally forming the monolithic body comprises additively manufacturing the monolithic body along a build direction parallel to the flow direction along the fluid flow path.
3. The method of claim 2, wherein integrally forming the monolithic body comprises forming the plurality of partial height fins along the build direction to include a concave leading edge and a convex trailing edge.
4. The method of claim 2, wherein integrally forming the monolithic body further includes: forming a rib along an opposite surface from which the plurality of partial height fins extend; and forming a connecting foot coupling a partial height fin to the rib, wherein the connecting foot maintains a shape of the partial height fin during manufacturing.
5. The method of claim 4, wherein integrally forming the monolithic body comprises forming the plurality of partial height fins along the build direction to include a concave leading edge and a convex trailing edge, wherein the connecting foot is adjacent the concave leading edge or the convex trailing edge.
6. The method of claim 5, wherein forming the connecting foot includes forming a first connecting foot adjacent the concave leading edge and a second connecting foot adjacent the convex trailing edge.
7. The method of claim 2, wherein integrally forming the monolithic body further comprises

- forming a plurality of connector fins connecting a first fin in a first row to a second fin in an adjacent row, the plurality of connector fins being obliquely angled relative to the build direction.
- 8.** The method of claim 7, wherein forming the plurality of connector fins comprises forming a plurality of openings through the plurality of connector fins.
- 9.** The method of claim 8, wherein the plurality of openings includes an array of oblong openings.
- 10.** The method of claim 7, wherein the plurality of connector fins are spaced from an opposite substrate from which the plurality of connectors fins extend.
- 11.** A method of manufacturing a heat exchanger comprising: integrally forming a monolithic body by: forming a first substrate; forming a second substrate, wherein a first fluid flow path is defined between the first substrate and the second substrate; forming a third substrate, wherein a second fluid flow path is defined between the second substrate and the first substrate; and forming a plurality of partial height fins extending between the first substrate and the second substrate, the plurality of partial height fins comprising: a first type of partial height fin extending from the first substrate into the second fluid flow path; and a second type of partial height fin extending from the second substrate into the second fluid flow path, wherein the plurality of partial height fins comprise a plurality of rows of partial height fins arranged side-by-side consecutively along the second fluid flow path, wherein fins of each row are offset from fins of an adjacent row in a direction perpendicular to a flow direction along the second fluid flow path.
- 12.** The method of claim 11, wherein integrally forming the monolithic body comprises additively manufacturing the monolithic body along a build direction parallel to the flow direction along the second fluid flow path.
- 13.** The method of claim 12, wherein integrally forming the monolithic body further including forming a plurality of stiffening elements between the first substrate and the second substrate.
- 14.** The method of claim 13, wherein the plurality of stiffening elements are obliquely angled relative to the build direction.
- 15.** The method of claim 13, wherein the plurality of stiffening elements correspond to a juncture between each adjacent row.
- 16.** The method of claim 12, wherein integrally forming the monolithic body further comprises: forming a fourth substrate, wherein a third flow path is defined between the third substrate and the fourth substrate; and forming a plurality of stiffening elements between the third substrate and the fourth substrate, wherein the plurality of stiffening elements are obliquely angled relative to the build direction.
- 17.** The method of claim 12, wherein integrally forming the monolithic body further comprises forming a plurality of connector fins connecting a first fin in a first row to a second fin in an adjacent row, the plurality of connector fins being obliquely angled relative to the build direction.
- 18.** The method of claim 17, wherein forming the plurality of connector fins comprises forming a plurality of openings through the plurality of connector fins.
- 19.** The method of claim 18, wherein the plurality of openings includes an array of oblong openings.
- 20.** The method of claim 17, wherein the plurality of connector fins are spaced from an opposite substrate from which the plurality of connectors fins extend.
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